Lab 8: Inference for a Single Mean and Errors in Inference

DATASCI 100: Introduction to Statistical Inference

Spring 2025

Overview

This lab is organized into two parts. In Part 1, we use the `t.test()` function in R to perform inference—both hypothesis testing and confidence interval estimation—for a single population mean. In Part 2, we treat a data set as our entire population, draw repeated random samples from it, and evaluate how well our inferential methods perform in terms of Type I error rates and confidence interval coverage.

Data

End-of-semester course evaluations are often criticized as indicators of instructional quality because they may reflect biases such as course difficulty or the physical appearance of the instructor. This data set contains course evaluation information for 463 classes taught by 94 randomly selected professors at the University of Texas at Austin. The data are in `CourseEvals.csv`. Selected variables are described in Table 1.

Table 1: Selected variables in `CourseEvals.csv`

Variable	Description
`prof_id`	Professor ID
`class_id`	Class ID
`course_eval`	Average course evaluation (1 = very unsatisfactory, 5 = excellent)
`prof_eval`	Average professor evaluation (1 = very unsatisfactory, 5 = excellent)
`rank`	Rank of professor: teaching, tenure track, or tenured
`ethnicity`	Ethnicity: not minority or minority
`gender`	Gender: 1 = male, 2 = female
`language`	Language of school where professor was educated: English or non-English
`age`	Age of professor
`cls_perc_eval`	Percent of students in the class who completed the evaluation
`cls_did_eval`	Number of students who completed the evaluation
`cls_students`	Total number of students in the class
`cls_credits`	Credits: one credit or multi-credit
`bty_avg`	Average beauty score (1 = lowest, 10 = highest)

Note: Because one professor may teach multiple courses, the observations are not truly independent. More complex methods would be appropriate; for simplicity in this course, please treat observations as independent.

Part 1: Inference for a Single Mean

Loading the Data

```r
# Import course evaluations data
evals <- read_csv("CourseEvals.csv")
```

Exploring the Response Variable

Researchers are concerned that non-response bias may invalidate the evaluation results. The university administration claims an overall 80% response rate. We investigate the variable `cls_perc_eval`, which records the percentage of students in each class who completed an evaluation.

```r
# Summary statistics
summary(evals$cls_perc_eval)
sd(evals$cls_perc_eval)

# Histogram
ggplot(evals, aes(x = cls_perc_eval)) +
  geom_histogram(bins = 20)
```

Among 463 courses, the percent completion is left-skewed with a mean of approximately 74.4 and a standard deviation of approximately 16.8.

One-Sample t-Test

To formally assess whether the true average completion rate differs from 80%, we conduct a one-sample t-test. Our hypotheses are

$$H_0 : \mu = 80 \quad \text{vs.} \quad H_a : \mu \neq 80,$$

where μ is the true average percent completion across all courses. Although the data are left-skewed, the large sample size ($n = 463$) ensures that the sampling distribution of the mean is approximately normal, so conditions for valid inference are satisfied.

```r
# Two-sided one-sample t-test (default)
t.test(evals$cls_perc_eval, mu = 80)
```

The `t.test()` function returns the test statistic, degrees of freedom, p-value, and a 95% confidence interval. By default it tests a two-sided alternative and reports a 95% confidence interval.

The test statistic is $t = -7.2$ with 462 degrees of freedom. At the $\alpha = 0.05$ significance level we reject H_0 ($p < 0.001$) and conclude that the true mean percent completion is significantly lower than 80%. We are 95% confident that the true mean is between 72.9% and 76.0%, confirming that the university falls short of its claimed response rate.

Changing the Confidence Level

To obtain a 90% confidence interval, use the `conf.level` argument.

```r
# 90% confidence interval
t.test(evals$cls_perc_eval, mu = 80, conf.level = 0.90)
```

The value supplied to `conf.level` must be between 0 and 1.

One-Sided Alternative Hypotheses

To test $H_0 : \mu = 80$ against the one-sided alternative $H_a : \mu < 80$, add the `alternative` argument.

```
# One-sided test (less than)
t.test(evals$cls_perc_eval, mu = 80, alternative = "less")
```

When a one-sided alternative is specified, the confidence interval has a lower bound of $-\infty$ (for a "less than" alternative) or an upper bound of $+\infty$ (for a "greater than" alternative).

Using the t Distribution Directly

Just as `pnorm()` and `qnorm()` work for the normal distribution, `pt()` and `qt()` work for the t distribution.

Calculating p-values with `pt()`

Suppose a one-sample t-test with a two-sided alternative yields $t = -2$ with 50 degrees of freedom. The p-value is twice the lower-tail area:

```
# Two-sided p-value for t = -2, df = 50
2 * pt(-2, df = 50)
```

This returns approximately 0.0509. For a positive test statistic $t = 2$, the p-value uses the upper-tail area:

```
# Using complement
2 * (1 - pt(2, df = 50))

# Equivalently, using lower.tail = FALSE
2 * pt(2, df = 50, lower.tail = FALSE)
```

Both approaches yield the same result (0.0509) because the t distribution is symmetric.

Calculating t-scores for Confidence Intervals with `qt()`

To construct a confidence interval manually, we need the critical t-value. For a 95% confidence interval with 50 degrees of freedom, the critical value corresponds to the 2.5th percentile (i.e., $\alpha/2 = 0.025$):

```
# Critical t-value (lower tail)
qt(0.025, df = 50)

# Equivalently, the 97.5th percentile
qt(0.975, df = 50)
```

The absolute value of the result—approximately 2.01—is the t-score used in the margin of error formula $\bar{x} \pm t^* \cdot \frac{s}{\sqrt{n}}$.

Part 3: Errors in Inference

Overview

In this part, we treat the 463 courses in `CourseEvals.csv` as the *entire population* of interest. Because we know the true population mean, we can evaluate how often our inferential methods make errors when applied to random samples from this population.

- A **Type I error** occurs when we reject H_0 even though H_0 is true. With $\alpha = 0.05$, we target a Type I error rate of 5%.
- A **missed capture** occurs when a confidence interval fails to contain the true parameter. With a 95% confidence level, we target a coverage rate of 95%.

Required Functions

This section uses two pre-written R functions distributed on Canvas as `TestingFunctions.R`:

- `inference.means()` — randomly selects samples from a numerical variable and performs inference on each sample.
- `plot.ci()` — plots the confidence intervals produced by `inference.means()`.

Load these functions with `source()`:

```
# Load the helper functions
source("TestingFunctions.R")
```

Population Distribution

We again examine `cls_perc_eval`, whose true population mean is $\mu = 74.4$ and standard deviation is $\sigma = 16.8$.

```
# Population histogram
ggplot(evals, aes(x = cls_perc_eval)) +
  geom_histogram(bins = 20)

# True population parameters
mean(evals$cls_perc_eval)
sd(evals$cls_perc_eval)
```

Running the Simulation

The `inference.means()` function accepts four arguments:

- `variable` — the numerical variable of interest.
- `sample.size` — the sample size n for each draw.
- `alpha` — the significance level (used for both testing and interval estimation).
- `num.reps` — the number of random samples to draw.

We take 100 samples of size $n = 50$ at $\alpha = 0.05$:

```r
# Run simulation (100 samples of size 50)
sim1 <- inference.means(variable    = evals$cls_perc_eval,
                        sample.size = 50,
                        alpha       = 0.05,
                        num.reps    = 100)

# View the results
View(sim1)
```

The resulting data frame has 100 rows—one per sample—and seven columns described in Table 2.

Table 2: Columns returned by `inference.means()`

Column	Description
`samp.est`	Point estimate (sample mean) from the draw of size n
`test.stat`	t-statistic for $H_0 : \mu = 74.4$ vs. $H_a : \mu \neq 74.4$
`p.val`	p-value for the above test
`decision`	*Reject* H_0 if $p \leq \alpha$; otherwise *fail to reject*
`lcl`	Lower confidence limit
`ucl`	Upper confidence limit
`capture`	Whether the interval captured the true mean $\mu = 74.4$

Assessing Assumptions for Valid Inference

Before evaluating the simulation results, we verify three conditions.

1. **Random sample.** The `inference.means()` function randomly selects observations from the population, so this condition is satisfied by design.
2. **Independence.** Because observations are drawn randomly, they are independent of each other, satisfying this condition as well.
3. **Approximate normality of the sampling distribution.** Although the underlying variable is left-skewed, our sample size ($n = 50$) is large enough for the Central Limit Theorem to guarantee that the sampling distribution of $\bar{x}$ is approximately normal.

Because all three conditions are satisfied, we expect the test to commit Type I errors at roughly the targeted rate of 5% and the confidence intervals to capture the true mean roughly 95% of the time.

Examining Hypothesis Test Performance

```r
# Distribution of sample means (sampling distribution)
ggplot(sim1, aes(x = samp.est)) +
  geom_histogram(bins = 20) +
  labs(title = "Sample Means")

# Distribution of t-statistics
ggplot(sim1, aes(x = test.stat)) +
  geom_histogram(bins = 20) +
  labs(title = "t Test Statistics")
```

```r

# Distribution of p-values
ggplot(sim1, aes(x = p.val)) +
  geom_histogram(bins = 20) +
  labs(title = "p-values")

# Frequency table of decisions
table(sim1$decision)
```

When the sampling distribution assumption is satisfied, the t-statistics will be approximately t-distributed and the p-values will be approximately uniformly distributed across $[0, 1]$. Any p-value below $\alpha = 0.05$ leads to rejecting H_0—a Type I error, since the null hypothesis is true by construction.

Examining Confidence Interval Performance

```r
# Plot the confidence intervals
plot.ci(results = sim1, true.val = 74.4)

# Frequency table of captures
table(sim1$capture)
```

Intervals that appear in red in the plot do not contain the true mean $\mu = 74.4$; these represent errors in estimation. Because the same samples that produced extreme test statistics also produced shifted confidence intervals, the instances of Type I errors and missed captures will coincide.

Agreement Between the Hypothesis Test and Confidence Interval

For inference about a single mean, the hypothesis test decision and the confidence interval conclusion always agree: if we reject $H_0 : \mu = \mu_0$, then μ_0 falls outside the corresponding confidence interval, and vice versa. We can verify this with a contingency table.

```r
# Contingency table: capture vs. decision
table(sim1$capture, sim1$decision)
```

You should observe that the two off-diagonal cells are both zero: there are no samples where the test rejected H_0 but the interval still captured μ, and no samples where the test failed to reject but the interval missed μ.

Long-Run Performance

Results from 100 samples are suggestive but subject to sampling variability. Increasing the number of simulations to 10,000 gives a clearer picture of long-run behavior.

```r
# Run 10,000 simulations
sim1long <- inference.means(variable    = evals$cls_perc_eval,
                            sample.size = 50,
                            alpha       = 0.05,
                            num.reps    = 10000)

# Long-run Type I error rate
table(sim1long$decision)
```

```r

# Long-run confidence interval coverage
table(sim1long$capture)
```

With 10,000 repetitions, the observed Type I error rate and confidence interval coverage should both be very close to their targeted values of 5% and 95%, respectively. Small deviations remain because of simulation randomness, but systematic deviations—say, an error rate of 8% or a coverage of 88%—would indicate that the inferential method is not performing as expected.

Ideas for this lab were obtained from lab materials written for OpenIntro Statistics by Andrew Bray and Mine Çetinkaya-Rundel, and from the CDC's Youth Risk Behavior Surveillance System documentation.